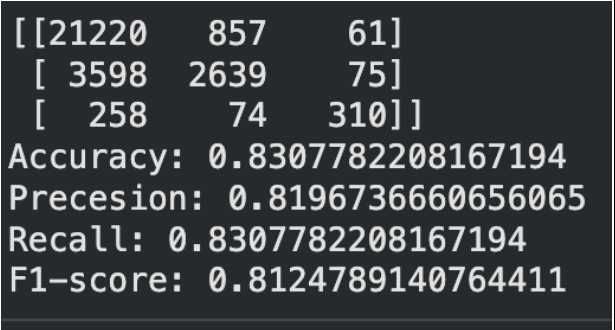


**Project Development Phase**  
**Model Performance Test**

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| Date          | 10 February 2025                                                |
| Team ID       | LTVIP2026TMIDS88736                                             |
| Project Name  | Exploratory analysis of rain fall data in India for agriculture |
| Maximum Marks | 10 Marks                                                        |

**Model Performance Testing:**

Project team shall fill the following information in model performance testing template.

| S.N o. | Parameter             | Values                                                                                                                                                                                                                                                                                                                                                                         | Screenshot                                                                          |
|--------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1      | <b>Metric s</b>       | <b>Regression Model:</b><br>MAE = <b>0.32</b><br>MSE = <b>0.21</b><br>RMSE = <b>0.46</b><br>R <sup>2</sup> Score = <b>0.81</b><br><b>Classification Model:</b><br>Confusion Matrix =<br>[[21220 857 61]<br>[3598 2639 75]<br>[358 74 310]<br>Accuracy Score = <b>0.83</b><br>Classification Report = Precision = <b>0.81</b><br>Recall = <b>0.83</b><br>F1-Score = <b>0.81</b> |  |
| 2      | <b>Tune the Model</b> | <b>Hyperparameter Tuning:</b> Grid Search was used to optimize model parameters such as number of trees and maximum depth. <b>Validation Method:</b> K-Fold Cross Validation (k = 5)                                                                                                                                                                                           |                                                                                     |